S.NO	DATE	TITLE	PAGE.NO	REMARK/ VIVA MARK	SIGNATURE

Exp.No: 1	Sequence to obtain user inputs display them using a message box
Date:	

Aim:

To create a sequence to obtain user inputs and display them using a message box.

Software Required:

- UiPath Studio
- PC with Windows 11

Procedure:

1. Create Variables:

- Define variables to store user inputs (e.g., `userInput1`, `userInput2`).

2. Use Input Dialog Activity:

- Drag and drop the **Input Dialog** activity into the sequence.
- This activity is used to collect input from the user.

3. Configure Input Dialog:

- Set the **Title** and **Label** to prompt the user.
- Store the entered value in the created variables using the **Result** property.

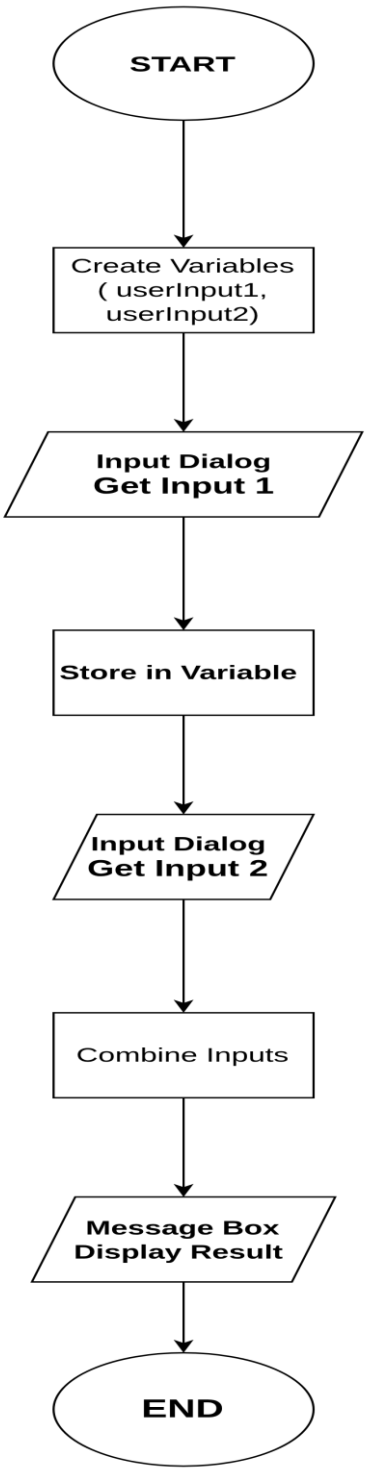
4. Use Message Box Activity:

- Drag and drop the **Message Box** activity into the sequence.

5. Display Inputs:

- Use the Message Box to display the collected inputs.
- Inputs can be concatenated or shown individually.

FLOWCHART:


START
Create Variables
(userInput1,
userInput2)
Input Dialog
Get Input 1
Store in Variable
Input Dialog
Get Input 2
Combine Inputs
Message Box
Display Result
END

Result:

A sequence to obtain user inputs and display them using a message box was successfully created.

Exp.No: 2	Flowchart to Navigate to a Desired Page Based on a Condition
Date:	

Aim:

To create a flowchart to navigate to a desired page based on a condition.

Software Required:

- UiPath Studio
- PC with Windows 11

Procedure:

1. Open UiPath Studio:

- Open UiPath Studio and create a new sequence.

2. Add Flowchart:

- Drag and drop a **Flowchart** from the Activities panel into the workflow designer.

3. Add Activities:

- Insert required activities inside the flowchart to perform conditional navigation.

4. Create Variable:

- Define a variable named **condition** of type Boolean.

5. Assign Activity:

- Use the **Assign** activity to set the value of the condition variable based on logic.

6. Decision Activity:

- Add a **Flow Decision** activity to check the condition.
- Connect it with the Assign activity.

7. If Condition is True:

- Add activities to navigate to **Page A** (e.g., using **Open Browser** or **Navigate To**).

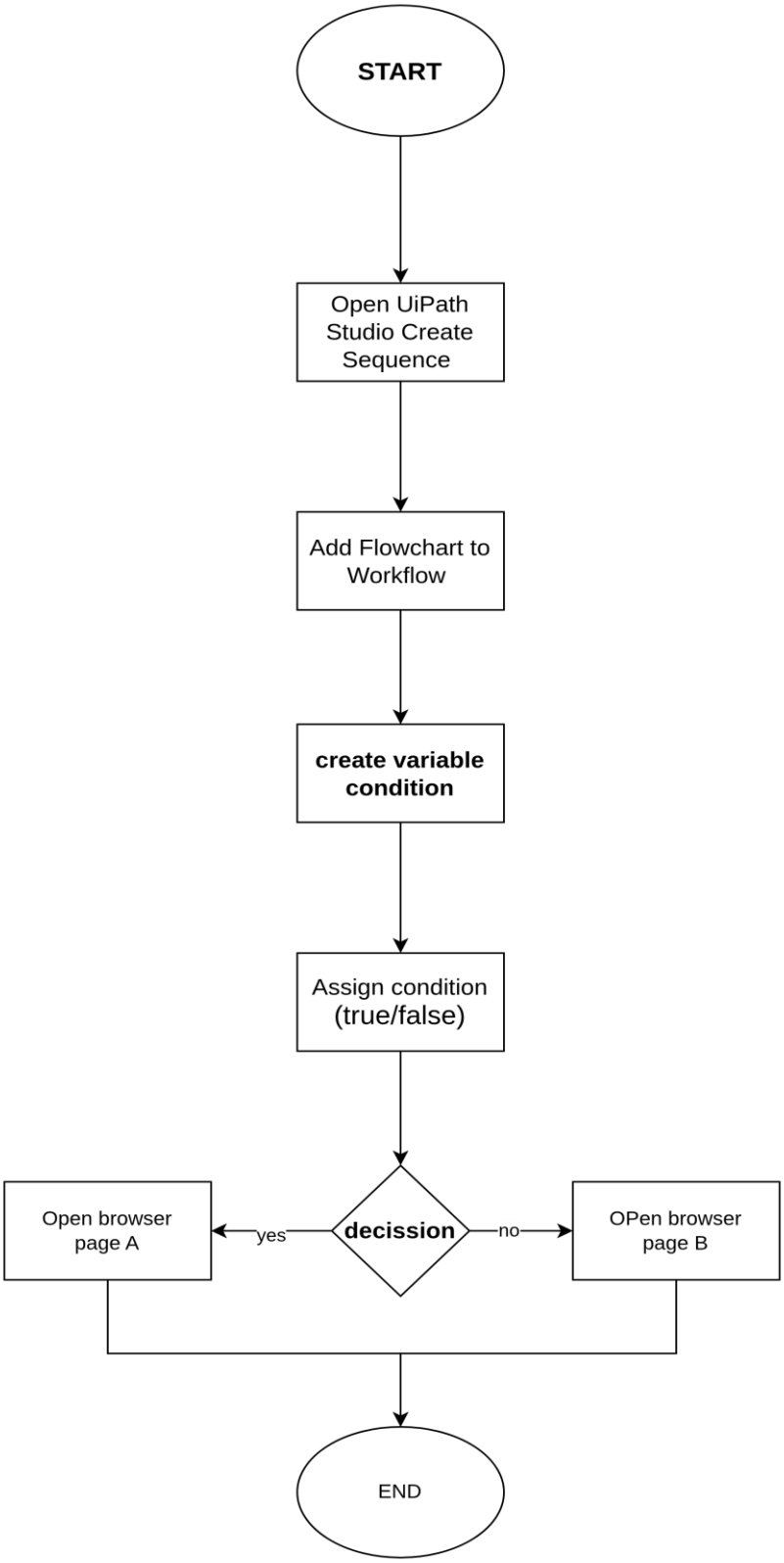
8. If Condition is False:

- Add activities to navigate to **Page B**.

9. Execution Flow:

- The flowchart will direct the process based on the evaluation of the condition.

FLOWCHART:


START
Open UiPath Studio Create Sequence
Add Flowchart to Workflow
create variable condition
Assign condition (true/false)
decission
yes
no
Open browser page A
OPen browser page B
END

Result:

A flowchart to navigate to a desired page based on a condition was successfully created.

Exp.No:3	State Machine Workflow
Date:	

Aim:

To create a State Machine workflow to compare user input with a random number.

Software Required:

- UiPath Studio
- PC with Windows 11

Procedure:

1. Create Project:

- Open UiPath Studio and create a new **State Machine** project.

2. Design States:

- Create the following states:
  - **Initialization State**
  - **Comparison State**
  - **End State**

3. Create Variables:

- `randomNumber` (Int32): Stores the generated random number
- `userInput` (String): Stores user input
- `isMatch` (Boolean): Stores comparison result

4. Initialization State:

- Use **Generate Random Number** activity to generate a number and store it in `randomNumber`.
- Use **Input Dialog** activity to get input from the user and store it in `userInput`.
- Transition to **Comparison State**.

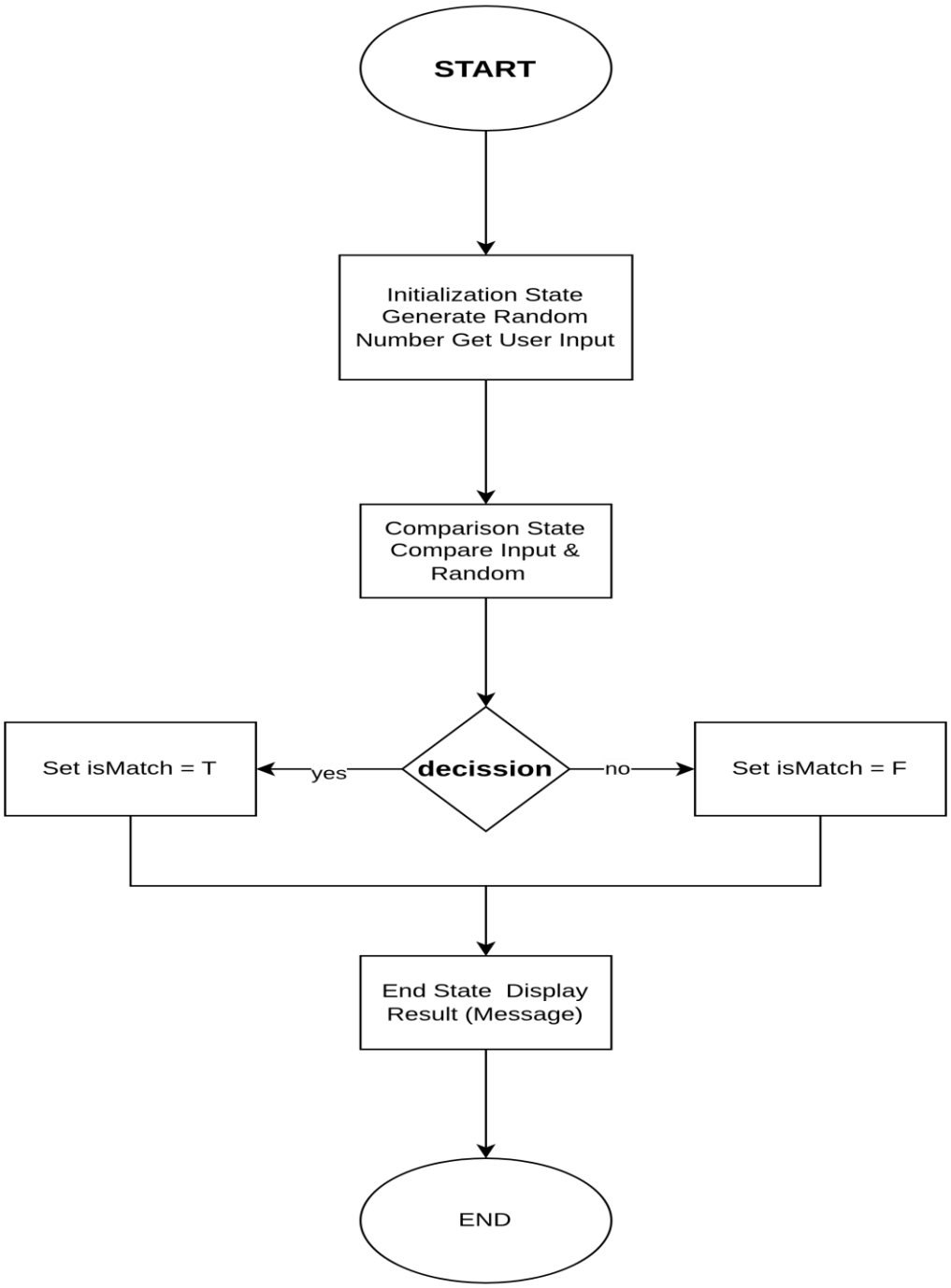
5. Comparison State:

- Use a **Flow Decision** or condition to compare:
  `randomNumber.ToString = userInput`
- If matched → set `isMatch = True`
- Else → set `isMatch = False`
- Transition to **End State**.

6. End State:

- Use a **Message Box** to display whether the user input matches the random number or not.

FLOWCHART:


START
Initialization State Generate Random Number Get User Input
Comparison State Compare Input & Random
decission
yes
no
Set isMatch = T
Set isMatch = F
End State Display Result (Message)
END

Result:

In this experiment, A State Machine workflow to compare user input with a random number was successfully created.

Exp.No: 4	Build a Process in the RPA Platform
Date:	

Aim:

To build a process in the RPA platform using UI Automation activities in UiPath.

Software Required:

- UiPath Studio
- PC with Windows 11

Procedure:

1. Create a New Sequence:

- Open UiPath Studio and click on **New Sequence** to create a workflow.

2. Add Activities to the Sequence:

a. Open Browser Activity:

- Drag and drop the **Open Browser** activity.
- Configure it to open a web browser (Chrome/Firefox) and navigate to a specific URL.

b. Attach Browser Activity:

- Drag and drop the **Attach Browser** activity.
- This defines the scope where actions will be performed.

c. Type Into Activity:

- Add the **Type Into** activity inside the Attach Browser.
- Indicate the input field (e.g., search bar) and enter text.

d. Click Activity:

- Add the **Click** activity inside the Attach Browser.
- Select the button or link to perform a click action.

e. Message Box Activity (Optional):

- Add a **Message Box** outside the browser scope to display a completion message.

3. Configure Activities:

- Set the URL in **Open Browser**.
- Provide input text in **Type Into**.
- Select the correct UI element in **Click**.

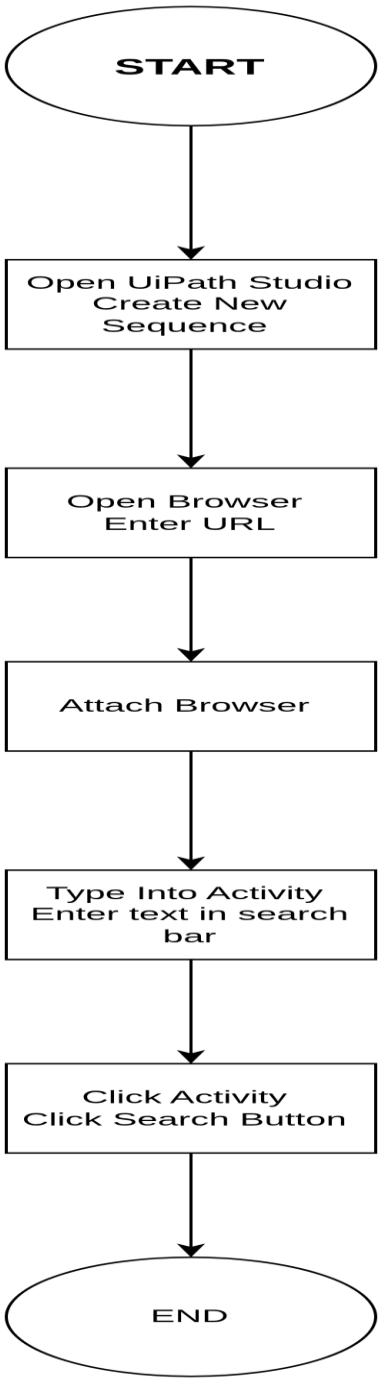
4. Adjust Selectors:

- Use **UiExplorer** to refine selectors if needed for accurate automation.

5. Run the Process:

- Save and execute the workflow.
- The automation will open the browser and perform the defined actions.

FLOWCHART:



Result:

In this experiment, A process in the RPA platform using UI Automation activities was successfully built in UiPath.

Exp.No: 5	Automation Process using System Activities, Variables & Arguments.
Date:	

Aim:

To create an automation process using key System Activities, Variables, and Arguments.

Software Required:

- UiPath Studio
- PC with Windows 11

Procedure:

1. Create a New Sequence:

- Open UiPath Studio and create a new sequence by clicking on **New Sequence**.

2. Define Variables and Arguments:

a. Variables:

- `length` (Double): Stores the length of the rectangle
- `width` (Double): Stores the width of the rectangle

b. Arguments:

- `result` (Double, Direction: Out): Stores the calculated area
- `errorMsg` (String, Direction: Out): Stores error messages

3. Add Activities to the Sequence:

a. Input Dialog Activity:

- Use **Input Dialog** to get user input for length and width.
- Store values in the variables.

b. Assign Activity:

- Use **Assign** to calculate area:
  `result = length * width`

c. Flow Decision Activity:

- Add **Flow Decision** to check if inputs are valid (greater than 0).
- If invalid, assign an error message to `errorMsg`.

d. Message Box Activity:

- Display either the calculated area or the error message.

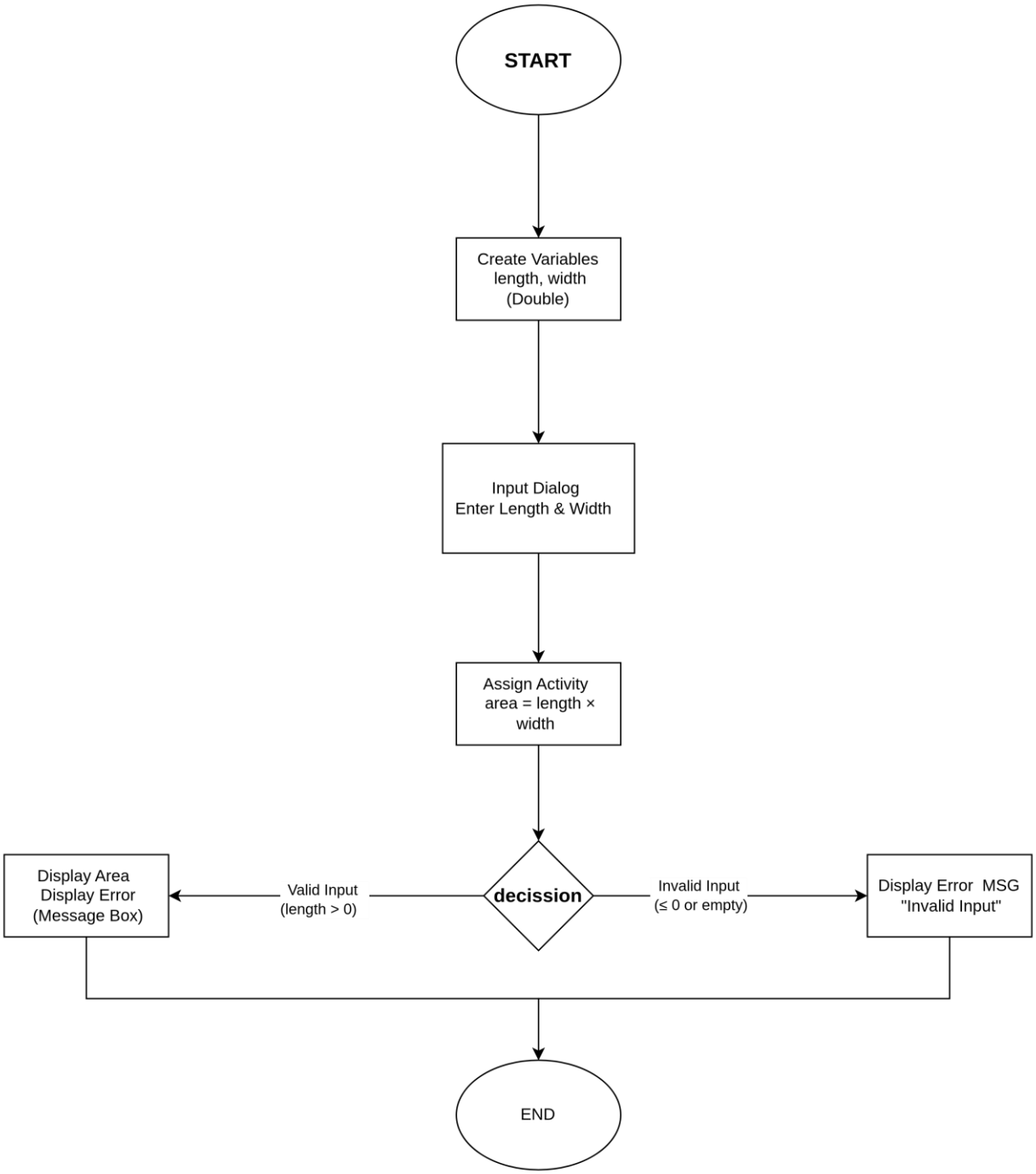
4. Configure Logic:

- Ensure correct expressions are used.
- Validate that length and width are positive values.

5. Run the Process:

- Save and run the workflow to execute the automation.

FLOWCHART:



Result:

In this experiment, An automation process using System Activities, Variables, and Arguments was successfully created in UiPath.

Exp.No: 6	Implement Automation using System Trigger
Date:	

Aim:

To implement automation using a system trigger.

Software Required:

- UiPath Studio
- PC with Windows 11

Procedure:

1. Create a New Project in UiPath Studio:

- Open UiPath Studio and create a new process or sequence based on your automation requirement.

2. Define Automation Steps:

- Add the required activities in the workflow to perform the automation tasks.

3. Publish the Project:

- Publish the project to UiPath Orchestrator so that it can be scheduled and triggered.

4. Configure a Trigger:

- In UiPath Assistant:
  - Open UiPath Assistant
  - Go to the **Automations** tab
  - Click **Create** and select the process
  - Choose **System Trigger** and define conditions (e.g., system startup, user login)
- In UiPath Orchestrator:
  - Log in to Orchestrator
  - Go to **Processes**
  - Select your process
  - Create a new trigger under the **Triggers** tab
  - Set trigger type as **System** and define conditions

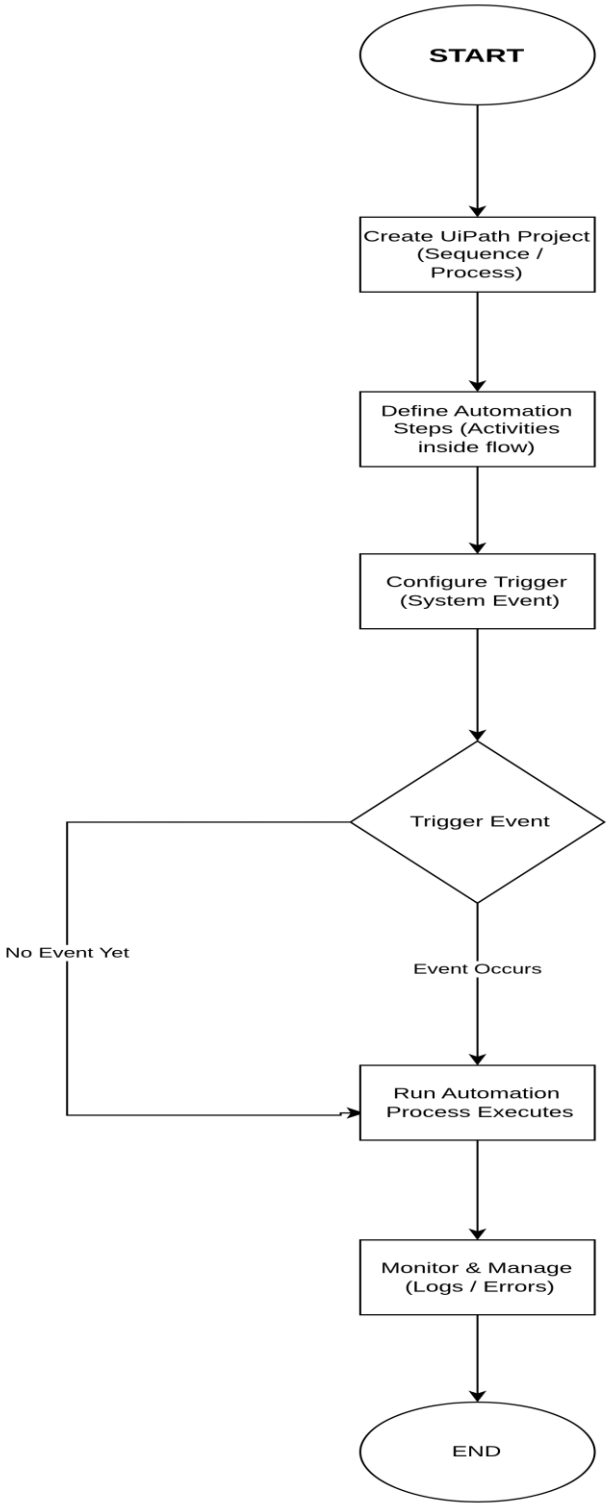
5. Test the Automation:

- Trigger the defined system event (e.g., system startup) and verify execution.

6. Monitor and Manage Automation:

- Check logs in UiPath Assistant or Orchestrator
- Modify triggers or workflow if needed

FLOWCHART:


START
Create UiPath Project (Sequence / Process)
Define Automation Steps (Activities inside flow)
Configure Trigger (System Event)
Trigger Event
No Event Yet
Event Occurs
Run Automation Process Executes
Monitor & Manage (Logs / Errors)
END

Result:

In this experiment, Automation using a system trigger was successfully implemented in UiPath Studio.

Exp.No: 7	Automate Web Email Login
Date:	

Aim:

To implement automation for automatic login to a web email account.

Software Required:

- UiPath Studio
- PC with Windows 11

Procedure:

1. Open UiPath Studio:

- Create a new sequence or open an existing workflow where you want to automate the login process.

2. Use 'Open Browser' Activity:

- Drag and drop the **Open Browser** activity.
- Configure it to open the browser and navigate to the email login page (e.g., https://mail.google.com).

3. Use 'Type Into' Activity:

- Enter the email address into the input field using **Type Into**.
- Indicate the field using UiExplorer or selector.

4. Use 'Click' Activity:

- Click the **Next** or **Continue** button after entering the email.

5. Use 'Type Secure Text' Activity:

- Enter the password securely using **Type Secure Text**.

6. Use 'Click' Activity:

- Click the **Sign In** button.
- Add delays or use **Wait for Element** activities to ensure proper loading.

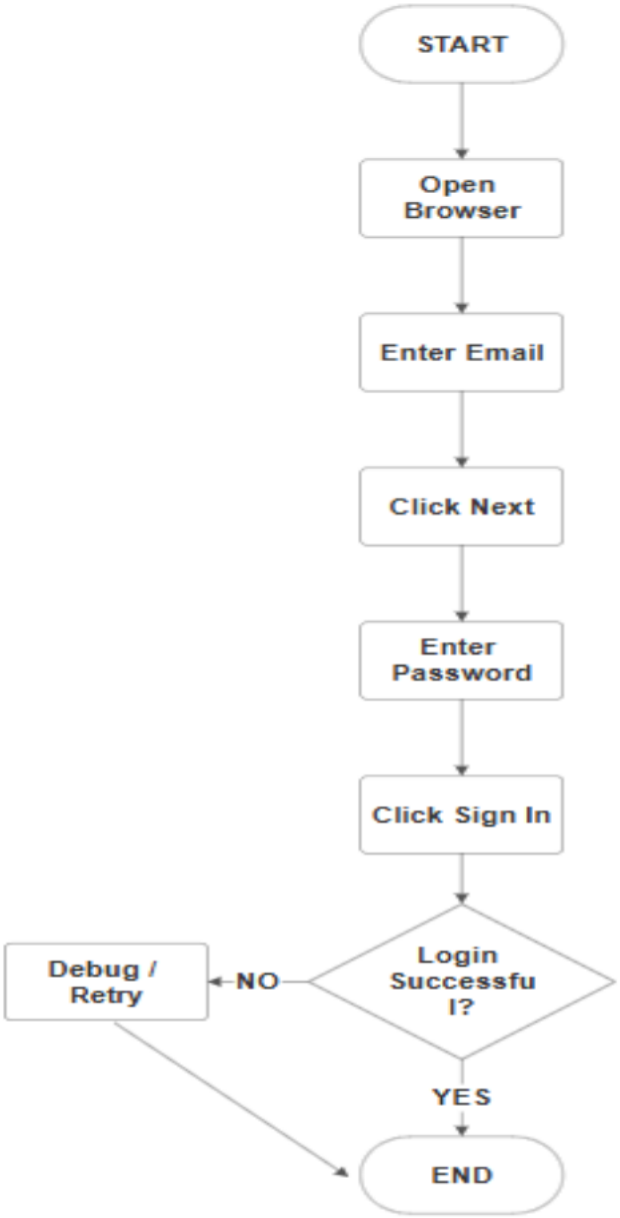
7. Additional Browser Activities:

- Use activities like **Click Text**, **Click Image**, or **Get Text** if required for further interaction.

8. Test and Debug:

- Run the workflow and fix issues by checking selectors and adjusting delays.

FLOWCHART


START
Open Browser
Enter Email
Click Next
Enter Password
Click Sign In
Login Successfu l?
NO
Debug / Retry
YES
END

Result:

In this experiment, Automation for recording mouse and keyboard actions was successfully implemented using UiPath Studio.

Exp.No: 8	Recording Mouse and Keyboard Actions
Date:	

Aim:

To implement automation for recording mouse and keyboard actions.

Software Required:

- UiPath Studio
- PC with Windows 11

Procedure:

1. Open UiPath Studio:

- Open UiPath Studio and create a new project or open an existing one.

2. Start Recording:

- Click on the **Recording** option in the ribbon menu.
- Choose:
  - **Basic Recording** – records actions in a specific application
  - **Desktop Recording** – records actions across the entire desktop

3. Perform Actions:

- Perform mouse clicks, keyboard inputs, and other actions on the screen.
- Interact with applications, fill forms, and click buttons as needed.

4. Stop Recording:

- After completing actions, click the **Stop** button in the recording toolbar (or use the assigned hotkey).

5. Generate Workflow:

- UiPath automatically creates a sequence based on recorded actions.
- The workflow contains activities corresponding to each action.

6. Edit and Validate:

- Review the workflow.
- Modify selectors, adjust delays, and add conditions or error handling if needed.

7. Run and Test:

- Execute the workflow to verify correct operation.

Note:

- Recorded workflows may require adjustments for different scenarios.
- Ensure selectors are reliable for stable automation.
- Modify recorded steps to improve performance and flexibility.

FLOWCHART:

START
Open UiPath Studio
Start Recording
Perform Actions
Stop Recording
Generate Workflow
Edit & Validate
Run & Test
END

Result:

In this experiment, Automation for scraping data from a website and writing it to a CSV file was successfully implemented using UiPath Studio.

Exp.No: 9	Web Scraping and Writing to CSV
Date:	

Aim:

To implement automation for scraping data from a website and writing it to a CSV file.

Software Required:

- UiPath Studio
- PC with Windows 11

Procedure:

1. Open UiPath Studio:

- Open UiPath Studio and create a new sequence or workflow.

2. Use Data Scraping:

- Drag and drop the **Data Scraping** activity into the workflow.
- Indicate the webpage or section from which data needs to be extracted.
- Use the Data Scraping Wizard to define the structure (columns, elements).
- Navigate through pages if required.
- The extracted data will be stored in a **DataTable** variable.

3. Use 'Write CSV' Activity:

- Drag and drop the **Write CSV** activity.
- Provide the **DataTable** as input.
- Specify the file path where the CSV file will be saved.

4. Run the Workflow:

- Execute the workflow to perform data scraping and save the results.

FLOWCHART

START
Open UiPath Studio
Data Scraping
Extract Data
Store in DataTable
Write CSV
Save File
Run Workflow
END

Result:

In this experiment, Error handling in the RPA platform was successfully implemented using UiPath Studio.

Exp.No:10	Error Handling in UiPath
Date:	

Aim:

To implement error handling in the RPA platform using UiPath.

Software Required:

- UiPath Studio
- PC with Windows 11

Procedure:

1. Use Try-Catch Activity:

- Add a **Try Catch** activity to handle exceptions.
- Place risky activities inside the **Try** block.
- Use the **Catch** block to handle errors (log, notify, retry, etc.).

2. Exception Handling Activities:

- **Throw Activity:** Used to generate exceptions manually.
- **Rethrow Activity:** Used to pass the exception to the parent workflow.

3. Logging:

- Use the **Log Message** activity to record information, warnings, or errors during execution.

4. Global Exception Handler:

- Configure a global exception handler in project settings.
- It handles unhandled errors by performing actions like logging or sending alerts.

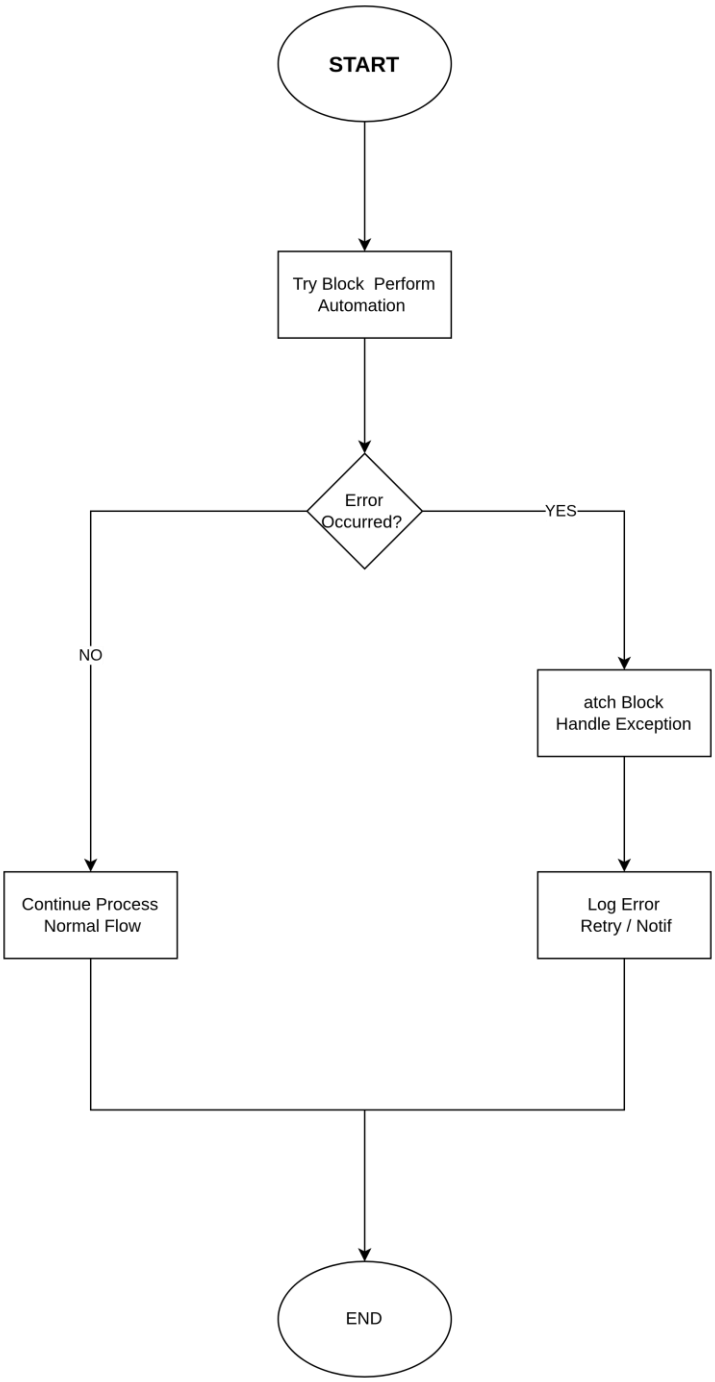
5. Retry Mechanism:

- Use **Retry Scope** activity to retry failed operations automatically.

6. Exception Types:

- Handle different exceptions (System or Business exceptions) appropriately based on their severity.

FLOWCHART:


START
Try Block Perform Automation
Error Occurred?
YES
NO
atch Block Handle Exception
Continue Process Normal Flow
Log Error Retry / Notif
END

RESULT:

In this experiment, Email query processing was successfully implemented in UiPath Studio.

Exp.No:11	Web Scraping
Date:	

Aim:

To implement web scraping in the RPA platform using UiPath.

Software Required:

- UiPath Studio
- PC with Windows 11

Procedure:

1. Using Web Scraping Wizard:

- Open UiPath Studio and create a new sequence or workflow.
- Drag and drop the **Data Scraping** activity.
- Indicate the webpage or section to extract data.
- Follow the wizard to select elements and define columns.
- Navigate through pages if required.
- Extracted data will be stored in a **DataTable** variable.

2. Using 'Extract Structured Data' Activity:

- Drag and drop the **Extract Structured Data** activity.
- Indicate the required webpage elements.
- Configure selectors and extraction options.
- Output will be stored in a **DataTable**.

3. Handling Dynamic Elements:

- Use delays, **Wait for Element**, or **Element Exists** activities to handle dynamically loaded content.

4. Validating Extracted Data:

- Check the DataTable to ensure data is accurate and complete.

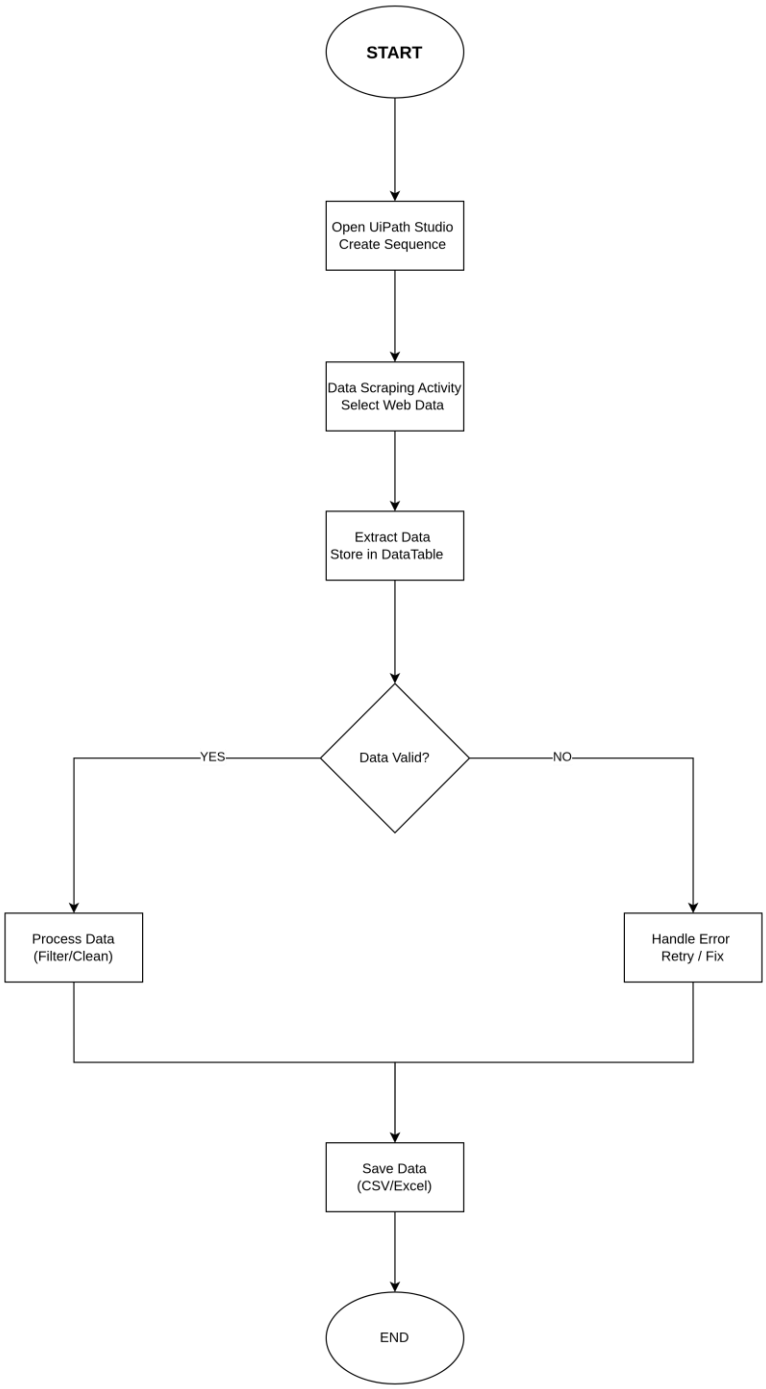
5. Processing Extracted Data:

- Perform operations like filtering, sorting, and cleaning.
- Save data to Excel, CSV, or databases.

6. Handling Errors and Exceptions:

- Use **Try Catch** to handle errors during scraping.
- Ensure workflow stability and reliability.

FLOWCHART:



Result:

In this experiment, Web scraping was successfully implemented in UiPath Studio.

Exp.No:12	Email Query Processing
Date:	

Aim:

To implement email query processing in the RPA platform using UiPath.

Software Required:

- UiPath Studio
- PC with Windows 11

Procedure:

1. Set up Email Activities:

- Install the UiPath.Mail.Activities package from Manage Packages.
- Use Get IMAP Mail Messages or Get POP3 Mail Messages to retrieve emails.
- Provide email credentials and server details.

2. Filter and Read Emails:

- Use a For Each loop to iterate through emails.
- Apply conditions based on subject, sender, or keywords.
- Use Save Attachments activity if needed.

3. Extract Information:

- Use string operations or regular expressions to extract required data from email content.

4. Take Actions Based on Information:

- Use If conditions to decide actions.
- Actions may include:
    - Sending replies
    - Updating databases
    - Triggering other workflows
    - Generating reports

5. Logging and Error Handling:

- Use Log Message activity to track execution.
- Use Try Catch to handle errors.

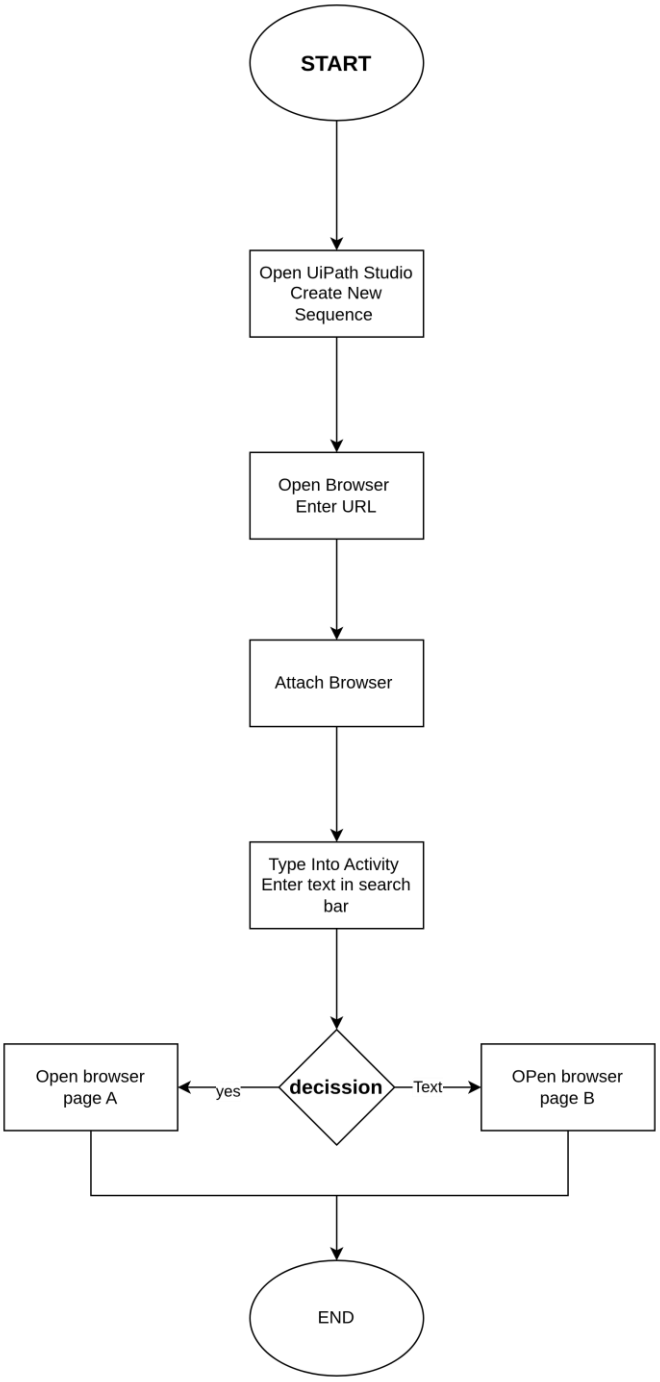
6. Testing and Validation:

- Test with different types of emails.
- Verify correct extraction and actions.

7. Security Considerations:

- Handle sensitive data securely.
- Follow best practices for data protection.

FLOWCHART:


START
Open UiPath Studio
Create New
Sequence
Open Browser
Enter URL
Attach Browser
Type Into Activity
Enter text in search
bar
Open browser
page A
yes
decission
Text
OPen browser
page B
END

Result:

In this experiment, Email query processing was successfully implemented in UiPath Studio.